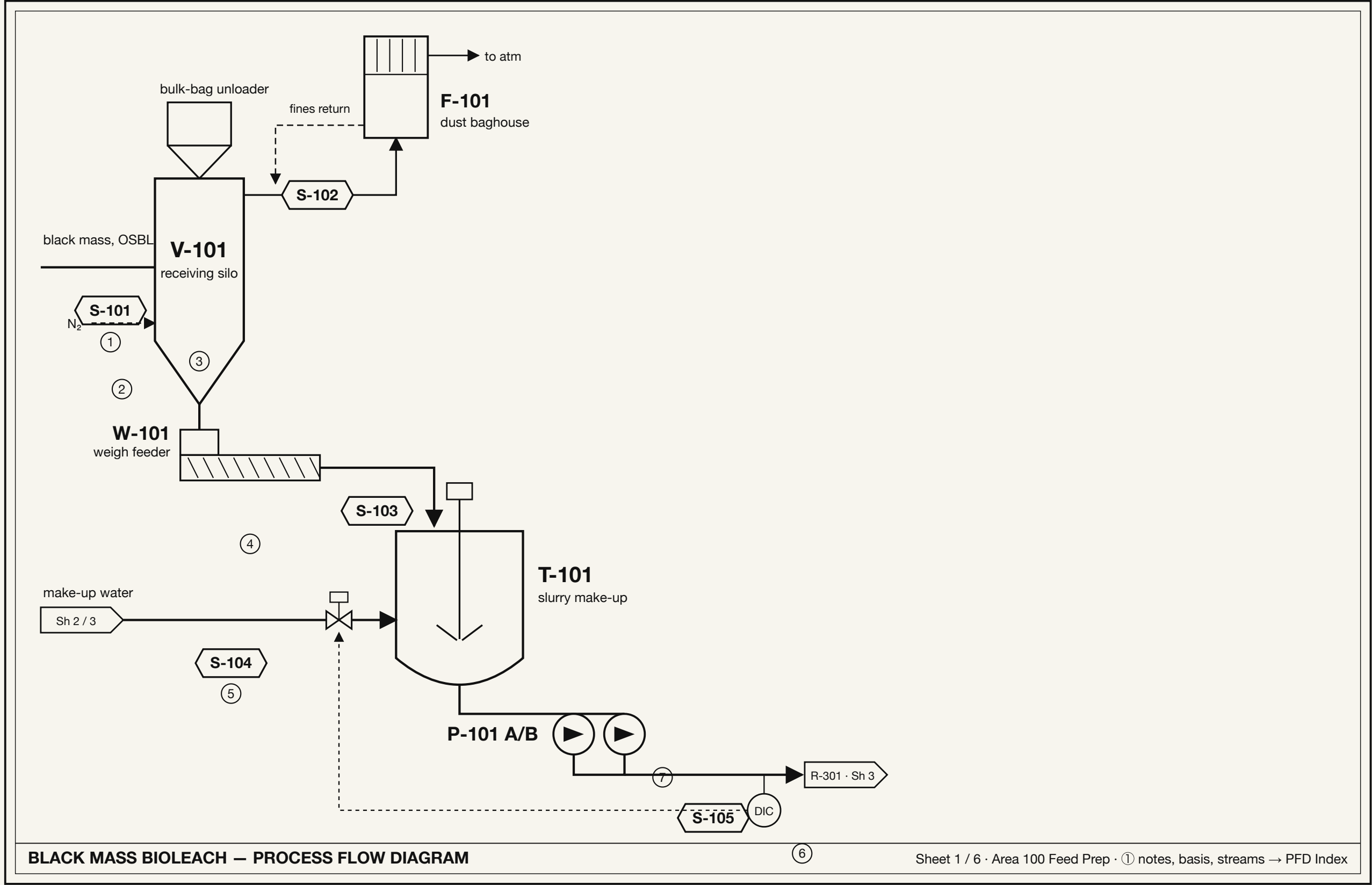


BLACK MASS BIOLEACH — PROCESS FLOW DIAGRAM

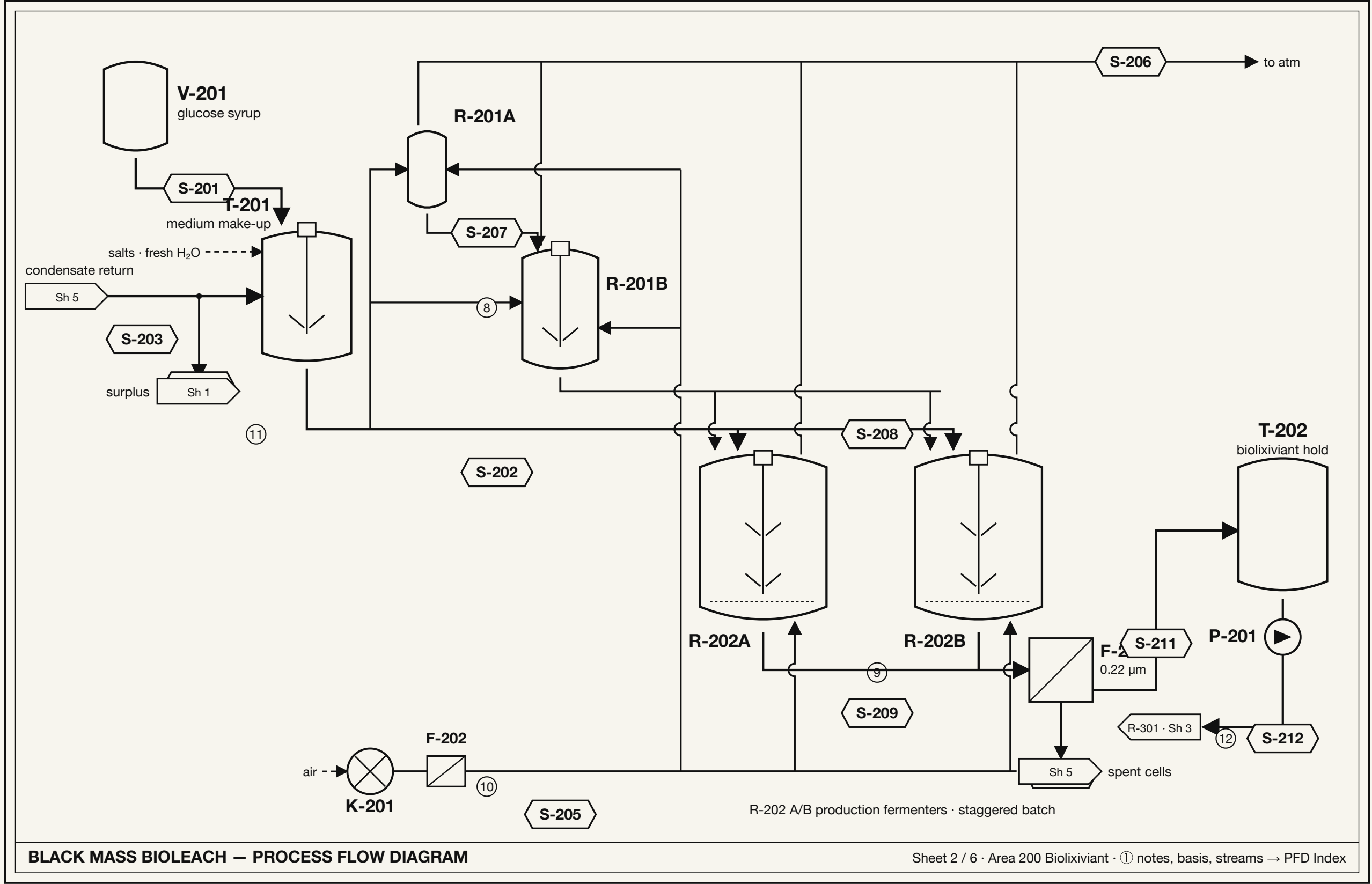
Zero-discharge refinery · 10,000 t/yr spent lithium-ion black mass · gluconic-acid biolixiviant

Sheets 1–6 combined. Equipment, sizing, stream tables and notes are in the companion PFD Index.

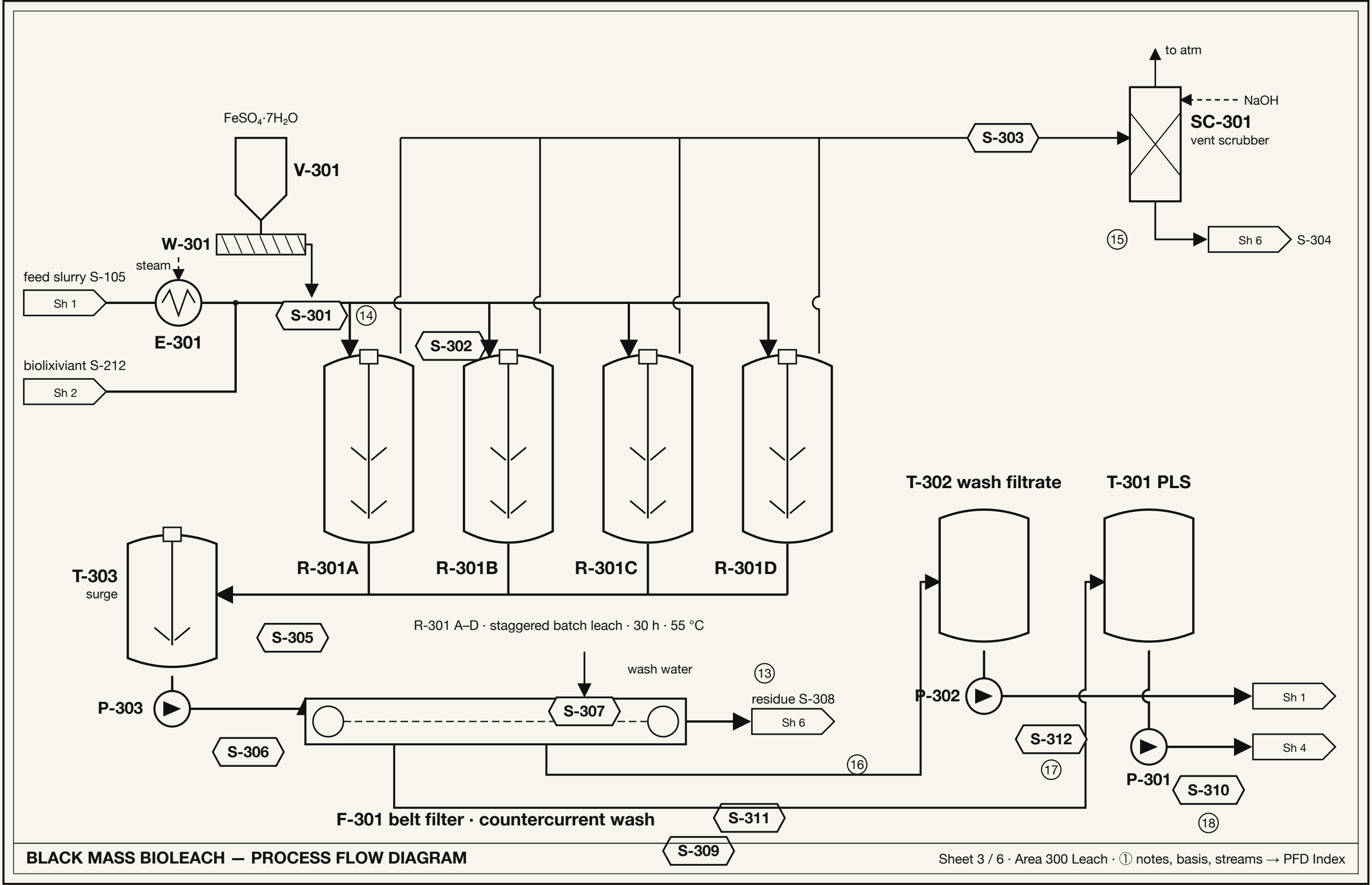
Sheet 1 — Area 100 · Feed preparation



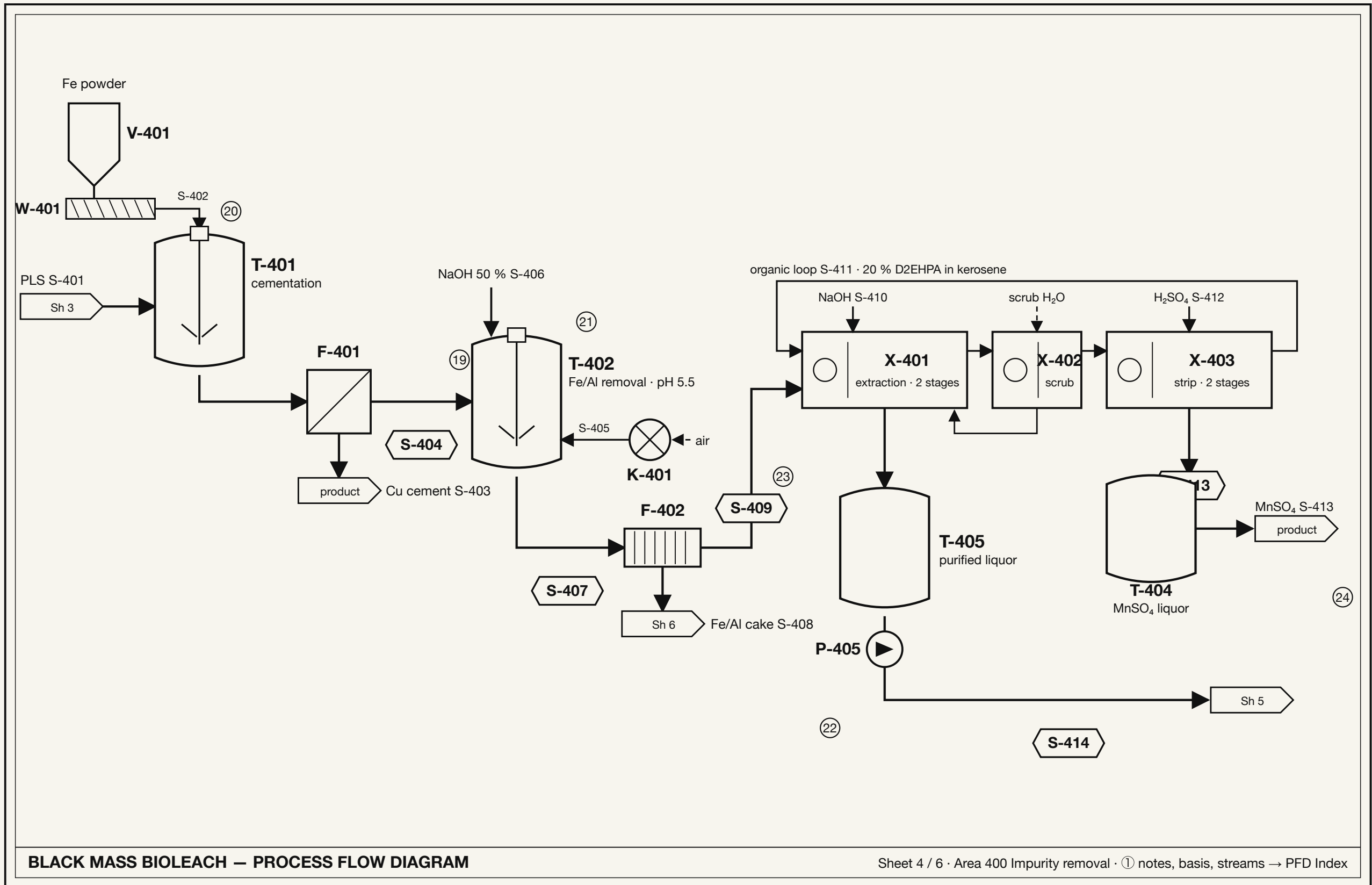
Sheet 2 — Area 200 · Biolixiviant fermentation



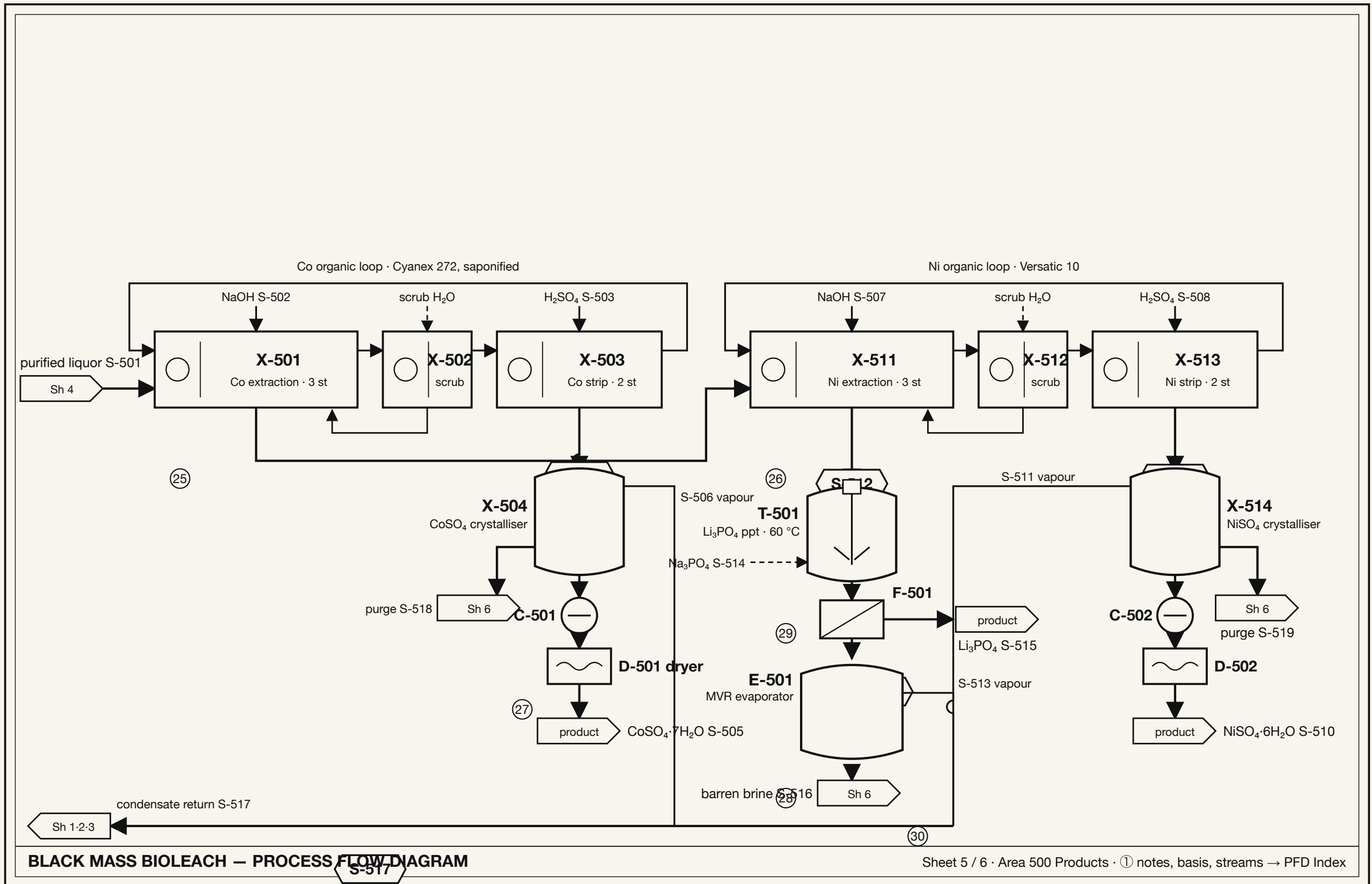
Sheet 3 — Area 300 · Batch leach and solid/liquid separation



Sheet 4 — Area 400 · Impurity removal (Cu → Fe/Al → Mn)



Sheet 5 — Area 500 · Products (Co SX → CoSO₄ · Ni SX → NiSO₄ · Li₃PO₄) and water return



Sheet 6 — Area 600 · Effluent, by-product salt and solid residues

